

Document Intelligence Platform

Intelligent Document Extraction, Validation & API Platform

AI Engineer Intern Technical Case Study

End-to-end document upload → extraction → validation → persistence → dashboard for Invoice, Balance Sheet, Profit & Loss and Cash Flow Statement documents.

Problem & Objective

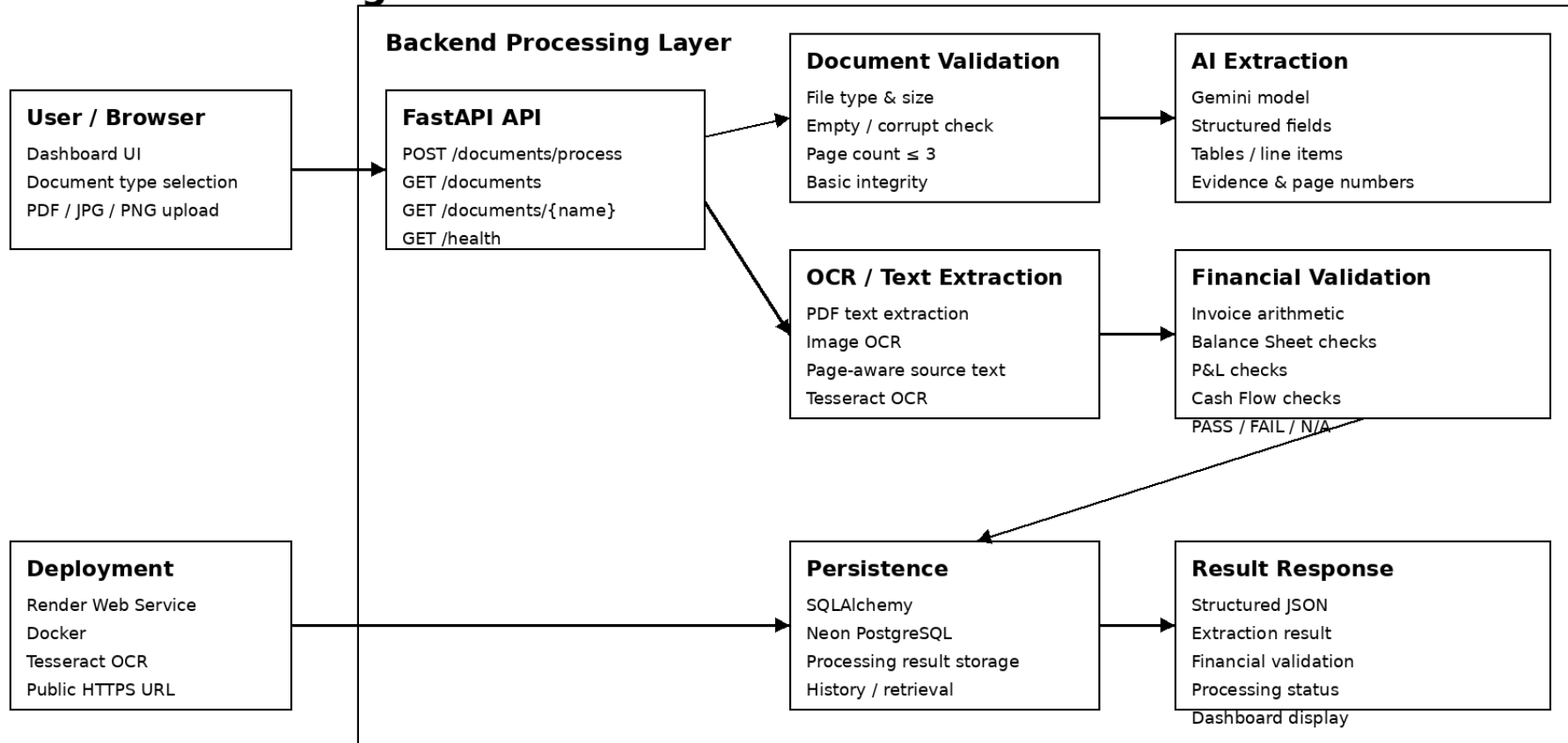
- Convert financial documents in PDF/JPG/PNG into structured information.
- Extract meaningful visible fields, comparative periods, tables and invoice line items.
- Validate financial relationships with deterministic arithmetic checks and tolerance.
- Expose REST APIs and a dashboard with persistent database storage.
- Return processing status plus evidence such as source text and page number.

Solution Architecture

- User/browser → FastAPI → validation → OCR/text extraction → Gemini structured extraction → financial validation → Neon PostgreSQL → JSON/dashboard.

Architecture Diagram

Document Intelligence Platform – Architecture



Technology Stack

- Python/FastAPI/Pydantic/SQLAlchemy
- PyMuPDF/Pillow/Tesseract OCR
- Google Gemini structured extraction
- Neon PostgreSQL
- HTML/CSS/JavaScript
- Render + Docker
- Swagger/OpenAPI

Production Dashboard — PASS

Document Intelligence Platform

Intelligent Document Extraction, Validation & API Platform

● API HEALTHY

Process Document

Upload a financial document for extraction and validation.

Document Type

Profit & Loss

▼

Document

Choose File

Consolidated Profit & Loss 2018.pdf

Supported formats: PDF, JPG, JPEG, PNG | Maximum 3 pages

Process Document

Document processed successfully.

Consolidated Profit & Loss 2018.pdf

Profit & Loss

PASS

Document Metadata

Structured Extraction & Evidence

Document Metadata

DOCUMENT TITLE Consolidated Statement of Profit and Loss	DOCUMENT DATE 2018-03-31	ENTITY HDFC Bank Limited	VENDOR -	CUSTOMER -	CURRENCY INR
PERIODS 31-Mar-18, 31-Mar-17					

Extracted Fields

Structured information extracted from the document.

Field	Value	Period	Evidence	Page
Interest earned	852,878,437	31-Mar-18	Interest earned 13 852,878,437	1
Interest earned	732,713,529	31-Mar-17	interest earned 732,713,529	1
Other income	160,566,041	31-Mar-18	Other income 14 160,566,041	1
Other income	128,776,329	31-Mar-17	Other income 14 128,776,329	1
Total Income	1,013,444,478	31-Mar-18	Total 1,013,444,478	1
Total Income	861,489,858	31-Mar-17	Total 861,489,858	1

Extracted Tables

Extracted Tables

Tables and line-item information extracted from the document.

Profit and Loss Statement

Particulars	31-Mar-18	31-Mar-17
Interest earned	852,878,437	732,713,529
Other income	160,566,041	128,776,329
Total Income	1,013,444,478	861,489,858
Interest expended	423,814,803	380,415,844
Operating expenses	239,272,220	207,510,707
Provisions and contingencies	164,749,045	120,689,285
Total Expenditure	827,836,068	708,615,836
Net profit for the year	185,608,410	152,874,022
Less: Minority interest	513,389	367,165
Add: Share in profits of associates	5,221	23,393
Consolidated profit for the year attributable to the Group	185,100,242	152,530,250
Impact on amalgamation [Refer Schedule 18(1)]	-	274,507

Financial Validation

Financial Validation

PASS

Arithmetic and reconciliation checks performed on extracted values.

Check	Formula	Calculated	Reported	Variance	Status
Profit and loss total income reconciliation - 31-Mar-18	Interest Earned + Other Income $\approx$ Total Income	1,01,34,44,478	1,01,34,44,478	0	PASS
Profit and loss total expenditure reconciliation - 31-Mar-18	Interest Expended + Operating Expenses + Provisions & Contingencies $\approx$ Total Expenditure	82,78,36,068	82,78,36,068	0	PASS
Profit after minority interest reconciliation - 31-Mar-18	Profit Before Minority Interest - Minority Interest $\approx$ Consolidated Net Profit	-	-	-	NOT_APPLICABLE
Profit and loss total income reconciliation - 31-Mar-17	Interest Earned + Other Income $\approx$ Total Income	86,14,89,858	86,14,89,858	0	PASS
Profit and loss total expenditure reconciliation - 31-Mar-17	Interest Expended + Operating Expenses + Provisions & Contingencies $\approx$ Total Expenditure	70,86,15,836	70,86,15,836	0	PASS
Profit after minority interest reconciliation - 31-Mar-17	Profit Before Minority Interest - Minority Interest $\approx$ Consolidated Net Profit	-	-	-	NOT_APPLICABLE

Processing History

Refresh

Previously processed documents stored in the database.

Persistent Processing History

Processing History

Refresh

Previously processed documents stored in the database.

Document	Type	Status	Processed At	Action
Consolidated Profit & Loss 2018.pdf	Profit & Loss	PASS	2026-09-11 20:36:11.025233	View

Validation Strategy

- Invoice: quantity $\times$ unit price, line totals, tax and total reconciliation.
- Balance Sheet: assets vs liabilities + equity and component sums.
- Profit & Loss: income/expenditure reconciliation for each period.
- Cash Flow: operating + investing + financing + FX and opening/closing cash.
- Missing required inputs produce NOT_APPLICABLE rather than assumptions.

Testing & Reliability

- Automated suite: 18 passed, 3 skipped after direct Gemini quota exhaustion.
- Production PASS flow verified with Consolidated Profit & Loss 2018.pdf.
- Production persistence and retrieval verified through Processing History and View.
- Gemini fallback handles model availability/quota failures.

Deployment & Security

- Public Render Web Service with Docker and Tesseract OCR.
- Neon PostgreSQL for persistent cloud storage.
- Secrets provided through environment variables and excluded from Git.
- Free-tier instances can sleep after inactivity, increasing first-request latency.

Limitations & Production Improvements

- External LLM availability and quotas affect extraction reliability.
- Three-page limit follows the case-study requirement.
- Production improvements: async job queues, stronger provider redundancy, retries/backoff, observability, authentication, rate limiting and object storage.

AI Coding Assistant Declaration

- ChatGPT was used for code generation, debugging, architecture guidance, testing support and documentation drafting.
- Generated code was reviewed, executed, tested and modified as required.
- The developer remains responsible for understanding and explaining the implementation.